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Practical No. 12

Aim: Open Ended Practical - Build an AI chatbot for FAQs.

Name of Student:

Roll No.:

Semester/Year: VI/III

Academic Session: 2025-26

Date of Performance:

Date of Submission:

AIM: Open Ended Practical - Build an AI chatbot for FAQs.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

- To design and implement a simple AI-based chatbot for answering frequently asked questions (FAQs).
- To understand the basic concepts of Natural Language Processing (NLP) and human-computer interaction.
- To learn how to use Python libraries like difflib for fuzzy matching and response generation. □ To simulate real-time conversation using input/output handling and typing effects.
- To develop problem-solving skills by creating logic for matching user queries with predefined responses.

Theory:

Concept: AI Chatbot for FAQs

An AI chatbot is a program that simulates human conversation to answer user queries. In this project, a rule-based chatbot is used, where predefined questions and answers are stored.

Working:

- The user enters a query, which is converted to lowercase.
- The chatbot checks for:
 1. Exact match
 2. Keyword match
 3. Fuzzy match using difflib
- The best matching response is returned.
- A typing effect is added to make interaction realistic.
- The program exits when the user types "exit", "quit", or "bye".

Technologies Used:

- Python
- difflib library
- Input/Output handling
- Time module

CODE:

```
from flask import Flask,
render_template, request, jsonify

import json

import re

from datetime import datetime

import random

app = Flask(__name__)

class AIChatbot:

    def __init__(self):

        self.faq_data = {

            "what is artificial
intelligence": "Artificial Intelligence
(AI) is a branch of computer science
that aims to create intelligent
machines that can simulate human
thinking, learning, and problem-
solving capabilities.",

            "types of ai": "There are
mainly three types of AI: 1) Narrow
AI (ANI) - designed for specific
tasks, 2) General AI (AGI) - human-
level intelligence, and 3)
Superintelligent AI (ASI) - exceeds
human intelligence.",

            "machine learning":
"Machine Learning is a subset of AI
that enables systems to learn and
improve from experience without
being explicitly programmed. It uses
algorithms to analyze data, identify
patterns, and make decisions.",

            "deep learning": "Deep
Learning is a subset of Machine
Learning that uses neural networks
with multiple layers to progressively
extract higher-level features from
raw input. It's particularly effective
for image recognition, natural
language processing, and complex
pattern recognition.",

            "neural networks": "Neural
Networks are computing systems
inspired by biological neural
networks. They consist of
interconnected nodes (neurons) that
process and transmit information,
```

```

enabling machines to learn from
data.",

def get_response(self, user_input):

    user_input =
user_input.lower().strip()

    # Check for greetings

    if any(word in user_input for
word in ['hello', 'hi', 'hey',
'greetings']):

        return
random.choice(self.greetings)

    # Check for FAQ matches

    for question, answer in
self.faq_data.items():

        if
self._match_question(user_input,
question):

            return answer

    # Check for partial matches

    for keyword in ['machine
learning', 'deep learning', 'neural
network', 'ai', 'artificial intelligence']:

        if keyword in user_input:

            related_questions = [q for q
in self.faq_data.keys() if keyword in
q]

            if related_questions:

                return f"I can help you
with {keyword}! Here are some
related topics: {'
'.join(related_questions[:3])}. Would
you like to know more about any of
these?"

            return
random.choice(self.fallback_respons
es)

    def _match_question(self,
user_input, faq_question):

        # Simple keyword matching for
FAQ

        user_words =
set(user_input.split())

        faq_words =
set(faq_question.split())

```

```

    # Check if significant overlap
    exists

    overlap =
    user_words.intersection(faq_words)
    if len(overlap) >= 2:
        return True

    # Check for partial matches
    for word in user_words:
        if any(word in faq_word or
        faq_word in word for faq_word in
        faq_words):
            return True

    return False

    def get_suggested_questions(self):
        return
    list(self.faq_data.keys())[:8]
chatbot = AIChatbot()
@app.route('/')
def index():
    return
    render_template('chatbot.html')

@app.route('/chat',
methods=['POST'])
def chat():
    user_message =
    request.json.get('message', '')
    if not user_message:
        return jsonify({'error': 'No
        message provided'}), 400

    response =
    chatbot.get_response(user_message)

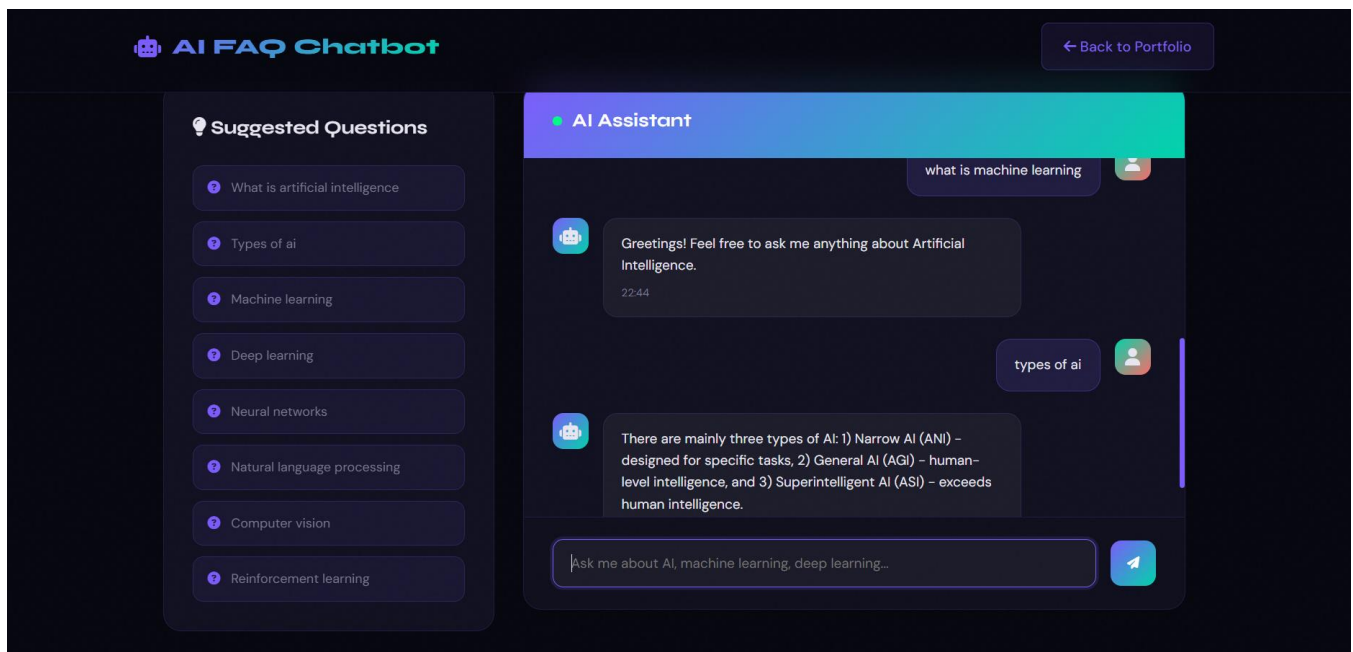
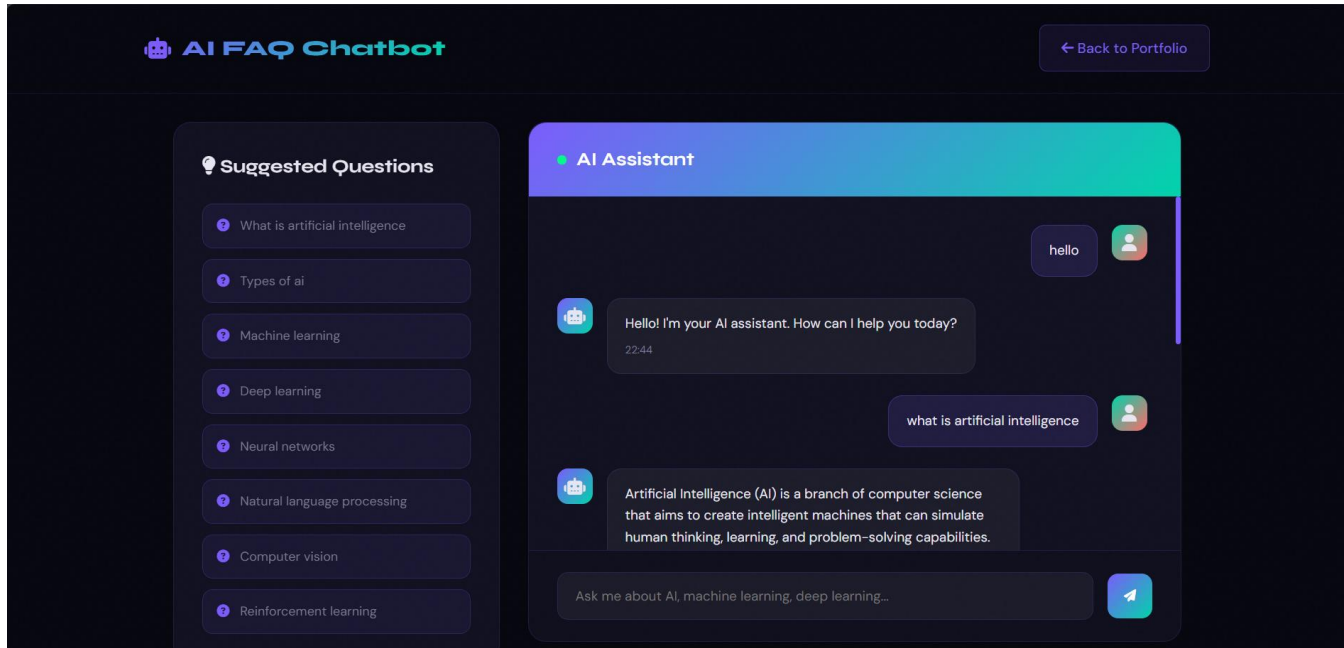
    return jsonify({
        'response': response,
        'timestamp':
        datetime.now().strftime('%H:%M'),
        'suggestions':
        chatbot.get_suggested_questions()
    })

@app.route('/suggestions')
def suggestions():

```

```
return jsonify({'suggestions':  
chatbot.get_suggested_questions()})  
  
if __name__ == '__main__':  
    app.run(debug=True,  
            host='0.0.0.0', port=5000)
```

SCREENSHOTS:



CONCLUSION:

The AI chatbot for FAQs was successfully developed using Python. It demonstrates how basic AI techniques like keyword matching and fuzzy logic can be used to simulate intelligent conversation. The project helps in understanding chatbot design, user interaction, and simple NLP concepts. Although the chatbot is rule-based, it provides a foundation for building more advanced AI systems using machine learning and deep learning in the future.

REFERENCES:

- Python Documentation – <https://docs.python.org/3/>
- difflib Library – <https://docs.python.org/3/library/difflib.html>
- IBM Cloud – Chatbot Overview – <https://www.ibm.com/topics/chatbots>
- GeeksforGeeks – Chatbot using Python – <https://www.geeksforgeeks.org>
- Towards Data Science – Introduction to Chatbots – <https://towardsdatascience.com>